

Interview Study Notes

Networking

OSI Model

- Please Do Not Throw Sausage Pizza Away

Application Layer Website, emails, apps

Presentation Layer Translate/Encrypt data

Session Layer Manage connection

Transport Layer Data come in right order

Network Layer Where data go

Data Link Layer Data Transfer

Physical Layer Hardware

Why is it important?

- It's easy to diagnose and fix problems using OSI model

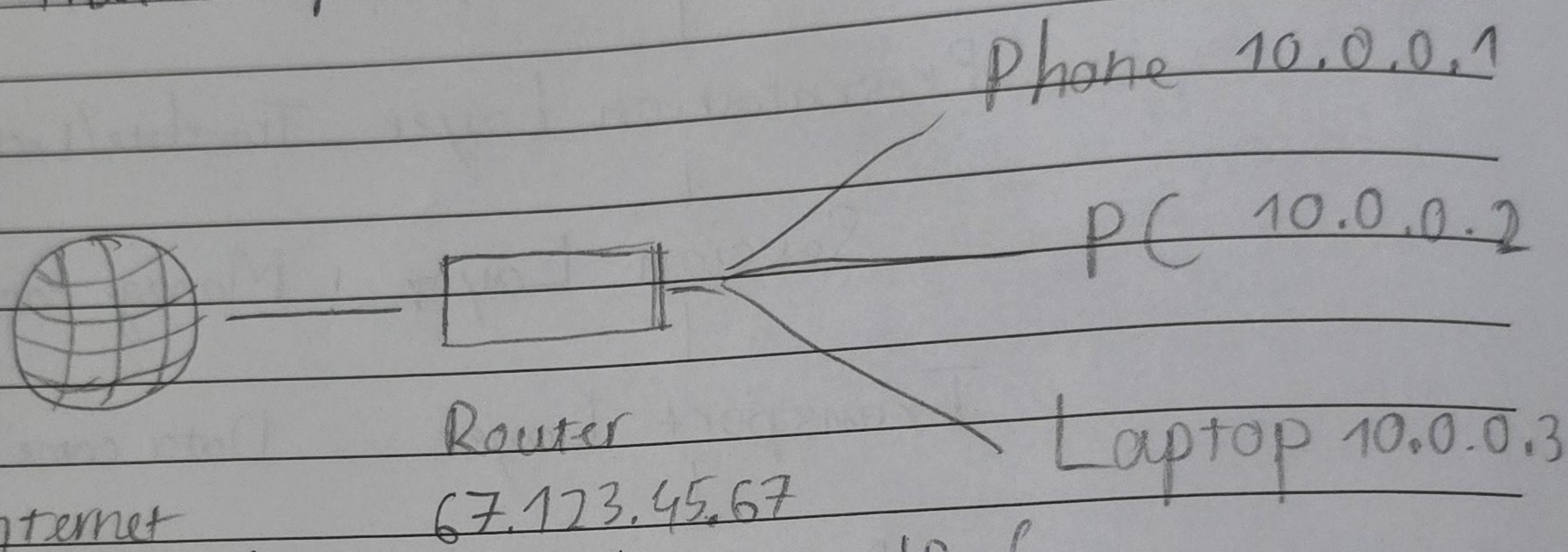
Firewall

- Allowing and Blocking network traffics
- Act as a safety barrier between private network & internet

Work by filtering incoming network data determine by rules in ACL (Access control list)

NAT (Network Address Translation)

- Used in Routers to translate a set of IP Address to another set
- There are limited number of IPv4 addresses ~ 4 billion
 - ↳ Not all devices can have public IP4 addresses
- NAT translate public to private IP Address and vice versa



Won't needed in future because IPv6

DNS (Domain Name System)

- Resolve name to numbers, domain names to IP address
- Yahoo.com → 74.125.44.25
- Work like a phone book

If computer can't find IP address in cache memory

↓ send query to Resolver (Internet Provider - Optus, Telstra, etc)

Resolver (if don't know) → Root Server 1.

↓ Authoritative Name Server 3.

TLD (Top Level Domain)

- Usually UDP (fast and lightweight) switch to TCP (secure but slower) if server send a redirect.

DNS Attacks

DNS Exfiltration:

- o Send data as sub-domains
 - o 268...abcd.badguy.com
 - o Doesn't show up in http logs because DNS traffic is less watched
- DNS sinkhole: a DNS server that intercepts known-bad domains and redirect them somewhere safe
 - DNS reads name right to left

DNS configs

- SOA - Start of Authority: main info (primary nameserver, admin contact, etc)
- A/AAAA - maps a domain to ~~IP~~ IPv4 (A) or IPv6 (AAAA)
- MX - SMTP mail exchangers, point to a mail server
- NS - Name Servers
- PTR - Pointer for Reverse DNS look up
- ~~DNS~~ CNAME - Domain name aliases

ARP (Address Resolution Protocol)

- Pair MAC address with IP address for IP connections
- Weak point - ARP spoofing - lying about which Mac own on IP

DHCP (Dynamic Host Configuration Protocol)

- Assign IP Addresses to hosts
- Also provide other network address (subnet mask, default gateway and DNS)
- Comes as client and server (67 for server, 68 for client)